Solutions / Answer Key

Sample topic: Unknown on both sides

60 minutes Mixed (easy to hard)

AI-assisted material. Review it before teaching.

Solutions

Worked solutions for every exercise, in the same order.

1. $3x + 7 = -23$

1. Subtract 7 from both sides:
$$3x = -30$$
2. Divide both sides by 3:
$$x = -10$$

Answer: $x = -10$

2. $4x - 4 = 24$

1. Add 4 to both sides:
$$4x = 28$$
2. Divide both sides by 4:
$$x = 7$$

Answer: $x = 7$

3. $-3x + 5 = 35$

1. Subtract 5 from both sides:
$$-3x = 30$$
2. Divide both sides by -3:
$$x = -10$$

Answer: $x = -10$

4. $-4x + 4 = -28$

1. Subtract 4 from both sides:
$$-4x = -32$$
2. Divide both sides by -4:
$$x = 8$$

Answer: $x = 8$

5. $2x - 8 = 8x - 11$

1. Subtract $8x$ from both sides:
$$-6x - 8 = -11$$
2. Add 8 to both sides:
$$-6x = -3$$
3. Divide both sides by -6:
$$x = \frac{1}{2}$$

Answer: $x = \frac{1}{2}$

6. $3x + 1 = 21$

1. Subtract 1 from both sides:
$$3x = 20$$
2. Divide both sides by 3:
$$x = \frac{20}{3}$$

Answer: $x = \frac{20}{3}$

No curriculum alignment is claimed.